

Exercise 2.

1. SELECT DISTINCT department,
FROM students;

department
IT
HR
Finance

2. SELECT department,
AVG (age) AS avg-age
FROM Students
GROUP BY department;

department	avg-age
IT	20,5
HR	22
Finance	23

3.

SELECT department,
COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

department	student_count
IT	2
HR	2

4 SELECT *

FROM Students,

WHERE age BETWEEN 21 AND 23;

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	EVE	22	HR

5 SELECT *

FROM Students

WHERE department = 'IT' OR 'HR' ~~AND~~

AND age > 21 ;

Student_id	name	age	department
2	Bob	22	HR
5	EVE Diana	22	HR

~~6 SELECT department,
total_Credits~~

6 SELECT department,
Credits

Sum (Credits) AS total_Credits

FROM courses

7. SELECT *
FROM courses
WHERE ~~credits~~ NOT Credits

7. SELECT *
FROM courses
WHERE credits NOT 4 ;

Course-id	Course-name	department	Credits
101	SOL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
Credits
FROM courses
ORDER BY credits DESC,

Course-id	Course-name	credits
	IT	4
	IT	4
	IT	3
	HR	3
	Finance	2

Course-id	Course-name	Credits
102.	Python	4
103	Data Science	4
101	SOL Basics	3
105	Statistics	3
104	Excel	2

9. SELECT MAX (grade) AS max-grade , AND
~~MIN~~ MIN (grade) AS min-grade AND
~~AVG~~ AVG (grade) AS ~~avg~~ avg-grade
 FROM enrollments ~~for~~ ;

max-grade	min-grade	avg-grade
90	78	84,6

10. SELECT COUNT (~~enrollment~~^{enrollment}-id) AS enrollment-count
 FROM enrollments
 GROUP BY course_id ;

course_id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11. SELECT department
 SUM (salary) AS total-salary AND
 SUM (bonus) AS total-bonus
 FROM salaries ;

department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. SELECT department
FROM Salaries
WHERE AVG (Salary) AS avg-Salary > 55000;

department	avg-Salary
IT	61000
finance	70000

13. SELECT employee-id,
name,
Salary,
bonus
SUM (Salary + bonus) AS total-Compensation
~~FROM salaries~~ ~~AS~~ FROM Salaries
~~GROUP BY~~ dep.

13. SELECT employee-id,
name,
Salary,
bonus
SUM (Salary + bonus) AS total-compensation
FROM salaries
~~HAVING (Salary + bonus > 60000)~~
HAVING total-compensation > 60000;

employee-id	name	Salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. SELECT department

AVG (budget) AS avg-budget,

SUM (budget) AS total-budget

FROM projects

GROUP BY department.;

department	avg-budget	total-budget
IT	135000	270000
finance	80000	80000
Marketing	60000	60000
HR	50000	50000

15. SELECT *

FROM projects

WHERE budget BETWEEN 50000 AND 120000

HAVING department NOT marketing.

Project-id	project-name	department	budget
1	AI APP	IT	120000
2	Payroll System	finance	80000
3	Dashboard	IT	15
5	HR portal	HR	50000